AI Learns Scientific Taste: Training Models to Judge and Propose High-Impact Research

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Read at ToKnow.ai

OpenMOSS at Fudan University trained AI to develop scientific taste: the ability to judge which research ideas have high potential impact and to propose new ones. Their method, Reinforcement Learning from Community Feedback (RLCF), uses citation patterns as training signal. They built SciJudgeBench, 696,758 field- and time-matched paper pairs from 2.1 million arXiv publications. A "Scientific Judge" model predicts which paper in each pair will receive more citations, controlling for field, year, and venue so the signal reflects quality rather than popularity. A second model, "Scientific Thinker," uses the Judge as a reward signal to generate research ideas rated higher in impact than baselines. Scientific Judge outperforms GPT-5.2 and Gemini 3 Pro at predicting paper impact, generalizing to future years, unseen fields, and peer-review preferences. All models and data are openly released.

Most AI-for-science work focuses on running experiments or writing papers. This targets the upstream question: what should researchers work on? A lab could use Scientific Judge to screen proposals or prioritize grants. The approach is data-efficient, needing just 11,700 training samples and one round of fine-tuning. The paper earned 228 upvotes and #1 Paper of the Month on Hugging Face, with 263 GitHub stars in days.

Citations encode visibility and collaboration networks, not just quality. If AI optimizes for what the community rewards, it risks reinforcing mainstream consensus over the unconventional ideas that drive breakthroughs. As one commenter noted, most scientific breakthroughs come from stubborn people going against the crowd.

Read more: Without Wonder, LLMs Are Analysts, Not Thinkers

Sources:

- AI Can Learn Scientific Taste (arXiv paper)
- Hugging Face Paper Page
- GitHub Repository
- OpenMOSS Model Collection
- SciJudgeBench Dataset
